Dylan Spektor

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EDUCATION & HONORS

University of Colorado Boulder **Expected: Apr 2028**

B.S. Applied Mathematics & B.A. Computer Science

GPA: 4.0

Organizations: CU Quants, Engineering Honors Program, Tau Beta Pi, Quantum Scholars.

Relevant Coursework: Software Development, Algorithms, Neural Networks & Deep Learning, Advanced Data Science, Probability & Statistics, Applied Real Analysis, Numerical Methods. **Spring 2027:** Machine Learning independent study.

Colorado Early Colleges Douglas County North **May 2025**

A.S. Computer Science (Dual Enrollment at Arapahoe Community College) | 1580 SAT, National Merit Finalist

RESEARCH & QUANTITATIVE EXPERIENCE

Undergraduate Researcher, Programming Languages & Verification Group **Aug 2026 – Present**

Advised by Professor Bor-Yuh Evan Chang, CU Boulder

- Selected for a yearlong research apprenticeship contributing to Demanded Abstract Interpretation of higher-order languages; targeting second authorship on a planned November 2026 PLDI submission.
- Built a small abstract interpreter in Lean 4; supporting current paper with Control Flow Graph analysis.

CU Quants Fund **Oct 2025 – Present**

Quantitative Developer

- Researching and developing volatility forecasting methods for cryptocurrency market making, with applications to perpetual-futures hedging.
- Design and implement data pipelines over historical order-book data to support research into stablecoin spread modeling and market-making opportunities.
- Tested crypto lead-lag predictive signals using cross-correlation and transaction costs; prototyped convex capital allocation across pairs strategies.

SELECTED PROJECTS & RESEARCH

Spectral Bias in Neural PDE Solvers **Current**

- Investigating spectral bias in neural networks for the heat and wave equations through Fourier/Sturm–Liouville analysis and its relationship to Neural Tangent Kernel (NTK) theory.
- Extending the analysis to Physics-Informed Neural Networks (PINNs), studying spectral bias in PDE residuals and its relationship to solution error, with computational demonstrations of mitigation methods planned.

Virtrade **April 2026**

- Collaborated in an agile team to build a full-stack paper trading web application using Node.js/Express, Python/Flask microservices, and PostgreSQL. Integrated real-time financial market data APIs to render interactive candlestick charts, and built a responsive UI with Bootstrap 5 and session-based user authentication.

Treasury Portfolio Optimization Calculator **Oct 2026**

- Built a minimum-cost Treasury dedication optimizer matching liability cash flows with mixed-integer linear programming, integer bond lots, and rolling surplus cash.
- Implemented Treasury cash-flow schedules and accrued-interest/dirty-price calculations with QuantLib; CLI ingests market data and outputs optimal CUSIP/principal allocations.

Riemann Zeta Zeros & Random-Matrix Spectral Statistics **Apr 2026**

- Co-authored a complex-analysis report relating Riemann zeta zeros to GUE/CUE spectral statistics in random matrix theory and the Hilbert–Pólya conjecture; derived moment-based connections and outlined a contour-integration approach to zero-spacing.

External Quantum Efficiency Degradation in Perovskite Cells **Dec 2025**

- In an experimental collaboration with the National Laboratory of the Rockies, analyzed weekly 350–400 nm EQE measurements across 0, 5, and 10 deposition cycles using bootstrap resampling, linear/quadratic regression, and ANOVA; found statistically significant differences in degradation coefficients.

TECHNICAL SKILLS & MISCELLANEOUS INTERESTS

Programming & Data: Python (NumPy, pandas, SciPy, scikit-learn, statsmodels, QuantLib), functional programming, Lean 4

Statistics & ML: probability, statistics, machine learning, deep learning, predictive modeling, time-series analysis, regression, bootstrapping, hypothesis testing, cross-correlation, clustering, dimensionality reduction

Interests: Philosophy of language, psychoanalysis, underground music, German idealism, mixed martial arts